Structural rigidity in the Erdős–Graham two-set permutation problem

William Kasel

August 2026

Abstract

Call a permutation of a set of positive integers *admissible* if it contains no increasing or decreasing three-term arithmetic progression in position order. Davis, Entringer, Graham, and Simmons proved that the positive integers are not admissibly permutable, partitioned them into three admissibly permutable sets, and asked whether two sets suffice. We study the canonical dyadic candidate $S_{\mathrm{A}} = \bigcup_{k\geq 2,\ k \text{ even}}(2^{k-1}, 2^k]$. We establish several structural restrictions on admissible permutations, including an orbit obstruction, an asymmetric balance law, and the impossibility of placing all sufficiently large dyadic blocks as contiguous runs. For the main result, we reduce admissibility of S_{A} to feasibility of an order gadget on $(M, 2M]$: the order must avoid monotone three-term progressions and satisfy fifteen precedence constraints induced by the values 15 and 16. We then prove that a three-constraint core of this gadget is inconsistent for every $M \equiv 0 \pmod 8$, using zigzag propagation on arithmetic-progression ladders, a transfer lock, and mirror-flood induction. Since every relevant dyadic scale lies in this residue class, S_{A} admits no admissible permutation. Thus the canonical dyadic partition does not solve the two-set problem, which remains open. Independent SAT encodings and certificate checks provide auxiliary verification of the human-readable proof.

1 Introduction

Call a linear arrangement of a set of integers *admissible* if no three of its terms, read in order of position, form an increasing or a decreasing arithmetic progression (a *monotone 3-AP*). Every finite set of integers admits an admissible arrangement: place the odd values before the even values and recurse, using the fact that the first and third term of a 3-AP share a parity (stated for intervals in [1], which credits earlier notes of this observation; for arbitrary finite sets in [2]). For infinite sets the situation changes completely. Davis, Entringer, Graham and Simmons [1] proved that every permutation of $\mathbb{Z}^+$ (of order type ω) contains an increasing 3-AP — in particular a monotone one — so $\mathbb{Z}^+$ itself admits no admissible permutation. They also constructed a partition of $\mathbb{Z}^+$ into *three* sets, each admitting an admissible permutation, and asked (Concluding Remarks 3, p. 89) whether two sets suffice. Erdős and Graham repeated the question in [2] (p. 338), calling it "a very annoying question" (see also their monograph [3]); it is problem #197 of the Erdős problem collection [12], where it is stated for the natural numbers and carries a Lean formalisation [13] (see Remark 3 for the $0 \in \mathbb{N}_0$ nuance).

The modern literature around the problem concerns densities, longer progressions, and enumeration (Ho [11] recently proved that $\lim_n \theta(n)^{1/n}$ does not exist, where $\theta(n)$ counts the 3-AP-free permutations of $[n]$, answering a question of Landman and Robertson). LeSaulnier and Vijay [6] exhibited a set admitting an admissible permutation with upper density 1/2 and lower density 1/4; writing $\alpha_{\mathbb{N}}(3)$ (resp. $\beta_{\mathbb{N}}(3)$) for the supremum of upper (resp. lower) densities of such sets, this gives $\alpha_{\mathbb{N}}(3) \geq 1/2$ and $\beta_{\mathbb{N}}(3) \geq 1/4$, and they conjectured both bounds sharp. Geneson [10] recently disproved the upper-density conjecture, proving $\alpha_{\mathbb{N}}(3) \geq 2/3$. His witness is strikingly relevant here: it is a union of ratio-4 doubling blocks $[L \cdot 4^j + m,\, 2L \cdot 4^j]$ — an S_{A}-like octave pattern *with the bottom sliver of each block removed* (here $m = M_{k-1}$ in his notation, re-cased to avoid a collision with our scale parameter M) and only finitely many octaves per stage. Our main theorem shows the full dyadic blocks fail, and the mechanism of failure — the attack of the values 15 and 16 on every block's bottom sliver (Sections 5–6) — is concentrated in exactly the region Geneson's construction excises, so the two results are consistent and their mechanisms align. (We do not claim that removing precisely those slivers is necessary or sufficient; his construction also uses stage separation and rapid scale growth.) Permutations of $\mathbb{Z}$ and of subsets of $\mathbb{Z}^+$ avoiding longer monotone progressions were constructed by Geneson [7] and Adenwalla [8]. Hirose and Saito [9] — continuing a line begun by Ardal, Brown and Jungić [4] on AP-avoiding ("chaotic") orderings of $\mathbb{Q}$ and $\mathbb{R}$ — characterised the *order types* of total orders on $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ admitting no monotone 3-AP; from their standpoint the difficulty of #197 is that order type ω is forced, where avoidance is hardest.

A natural first candidate for a two-set partition is the *dyadic partition*: let

$$S_{\mathrm{A}} \;=\; \bigcup_{k \text{ even},\, k\geq 2} \left(2^{k-1},\, 2^{k}\right] \;=\; (2,4] \cup (8,16] \cup (32,64] \cup \cdots,$$

$$S_B \;=\; \mathbb{Z}^+ \setminus S_{\mathrm{A}} \;=\; \{1,2\} \cup (4,8] \cup (16,32] \cup \cdots,$$

so that each team is a union of doubling blocks (together, for S_B, with the two initial values) and the completion $2y - x$ of a within-block pair $x < y$, whenever it escapes the block upward, lands in the adjacent block — the other team's territory. Two elementary density facts explain why this is the canonical candidate. First, S_{A} has upper density 2/3 (attained along $n = 4^k$) and lower density 1/3 (along $n = 2 \cdot 4^k$), and S_B the same with the roles of the two subsequences exchanged. So S_{A} sits exactly at the current record value $\alpha_{\mathbb{N}}(3) \geq 2/3$ of [10], and the main theorem below therefore exhibits a specific set *at* that density which is not permutable — the record is not improved by taking full dyadic blocks. Second, for any partition $\mathbb{Z}^+ = A \sqcup B$ the identity $\overline{d}(A) + \underline{d}(B) = 1$ holds automatically, so a two-set solution requires $\alpha_{\mathbb{N}}(3) + \beta_{\mathbb{N}}(3) \geq 1$ (a remark of [6], who conjectured the sum to be $3/4 < 1$); the dyadic pair meets this constraint exactly, $2/3 + 1/3 = 1$, while the best bounds currently known give only $\alpha_{\mathbb{N}}(3) + \beta_{\mathbb{N}}(3) \geq 2/3 + 1/4 = 11/12$, so the density obstruction is undecided. Our main result eliminates this candidate on other grounds.

> **Main Theorem** (Theorem 21). *S_{A} admits no permutation of order type ω free of monotone 3-term arithmetic progressions. Consequently the dyadic partition does not witness a positive answer to problem #197.*

The proof chain is short and we state it first, since the paper contains a good deal of surrounding material that is not part of it. It has exactly two links. The first is a reduction

(Theorem 16): an *order gadget* $\mathrm{OG}(M)$ on the interval $(M, 2M]$ — AP-freeness plus fifteen precedence axioms induced by the two values 15 and 16 — whose infeasibility at infinitely many dyadic scales implies the main theorem unconditionally. It is proved directly on a hypothetical permutation, in three paragraphs, and uses nothing else in the paper. The second — and this is the mathematical heart — we prove by hand that already a three-axiom core of $\mathrm{OG}(M)$ is inconsistent with admissibility for every $M \equiv 0 \pmod 8$ (Theorem 22), via a toolkit of ladder lemmas whose single appeal to the residue of M occurs at two "flood centres" near $3M/2$. (The basic zigzag alternation on AP-ladders that seeds the toolkit is classical — it appears inside [1] in Folkman's argument, as Lemma 2.5 of [9], and as Lemmas 2.1–2.2 of [10]; what is new here, to our knowledge, is its boundary-quantitative use on bounded ladders, the transfer lock, and the flood induction.) Computationally, the mod-8 condition is sharp at every tested scale: the same three axioms are satisfiable at every tested scale with $M \not\equiv 0 \pmod 8$ (a parametric satisfying construction for the complementary residues is not proved here, and the main theorem does not require one), matching the machine's phase diagram exactly.

The paper is organised so that this chain stands alone. Section 3 collects four obstructions that constrain admissible permutations of any set — an orbit obstruction, an asymmetric balance law, a ratio-ascent lemma and a partition-rigidity lemma — and Section 4 proves that no admissible permutation of S_{A} can play all sufficiently large dyadic blocks as contiguous runs. Neither is used later; both are included because they are the general facts we know about the problem, and the last of them motivates the dyadic candidate. Sections 5 and 6 are the proof. Section 7 separates the three roles machine work plays here and records what is and is not certified. Two appendices hold the rest: Appendix A develops the *chunk calculus*, an exact reformulation of admissibility in terms of stage functions, which was the coordinate system of the search and which gives a second route to the reduction (Remark 17); and Appendix B records the finite-scale computations and the conjectures they suggest. Nothing in either appendix is used in the proof of Theorem 21.

A word on methodology. The elimination phase and the discovery of the gadget, the three-axiom core, and the residue phenomenon were carried out with large-scale SAT-solver experimentation (CaDiCaL, OR-Tools). The load-bearing negative claims — the infeasibility of the order gadget and of its three-axiom core — were cross-validated by independently written encodings and by more than one solver; the auxiliary sweeps reported in Appendix B were not, and are flagged where they occur. The final proof chain, however, does not rest on any solver verdict: the reductions and the C3 core theorem are human-readable arguments. They are $\forall M$ schemas, with infinitely many instances; a finite but large range of those instances (Remark 29) has been machine-audited step by step by three independently written checkers, including mutation testing of the checkers themselves — an audit of the writing, not a substitute for the proofs. A reproduction kit (scripts, pinned environments, and DRAT certificates for two base instances) accompanies the paper.

Problem #197 itself remains open. Beyond eliminating the leading candidate, the methods here — the chunk calculus, the attack/guard analysis of small values against doubling blocks, and the residue-locked ladder toolkit — should apply to any two-set partition whose parts are unions of intervals, and we expect them to bear on the general question.

2 Definitions and conventions

Definition 1. A set $S \subseteq \mathbb{Z}^+$ is *permutable* if there is a bijection $\pi\colon \mathbb{N} \to S$ such that no $i < j < k$ have $\pi(i), \pi(j), \pi(k)$ forming an arithmetic progression with $\pi(i) < \pi(j) < \pi(k)$ or $\pi(i) > \pi(j) > \pi(k)$. When the progression length matters we write 3-*permutable*; in this paper the two words are synonymous, since only 3-term progressions are forbidden. Note that a permutable set is by definition countably infinite and the permutation has order type ω.

The completion of an ordered pair: if x is placed before y, the value $z = 2y - x$ (which continues the progression through x, y in the direction of y) must not be placed after y; equivalently every such $z \in S$ placed later yields a monotone 3-AP.

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$ is the set of positive integers, which serves both as the ground set (written $\mathbb{Z}^+$ when we wish to emphasise that role) and as the index set of positions in a permutation.

Definition 2 (Dyadic indexing)**.** For $k \geq 1$ the *k-th dyadic block* is

$$B_k \;=\; \Bigl(2^{k-1},\, 2^k\Bigr] \cap \mathbb{Z}^+ \;=\; \{\, 2^{k-1}+1,\; 2^{k-1}+2,\; \dots,\; 2^k \,\}, \qquad |B_k| = 2^{k-1}.$$

The blocks $B_1, B_2, B_3, \dots$ are pairwise disjoint and partition $\mathbb{Z}^+ \setminus \{1\}$; for $v \geq 2$ we write $\operatorname{block}(v)$ for the unique $k \geq 1$ with $v \in B_k$. The canonical two-set candidate is

$$S_{\mathrm{A}} \;=\; \bigcup_{k \geq 2,\; k \text{ even}} B_k \;=\; (2,4] \cup (8,16] \cup (32,64] \cup \cdots, \qquad S_B \;=\; \mathbb{Z}^+ \setminus S_{\mathrm{A}}.$$

The range $k \geq 2$ in the union is not a restriction, since $k = 1$ is odd; we nevertheless state it, because two counts used later depend on knowing exactly which small values belong to S_{A}: $1 \notin S_{\mathrm{A}}$ and $2 \notin S_{\mathrm{A}}$, so $S_{\mathrm{A}} \cap [1,16] = \{3,4\} \cup \{9,10,\dots,16\}$ has exactly ten elements, and $|S_{\mathrm{A}} \cap [1,64]| = 42$. Note that $1 \in S_B$ lies in no block and $\operatorname{block}(1)$ is undefined; we apply block only to elements of S_{A}, all of which are ≥ 3.

Remark 3 (Ground set: $\mathbb{Z}^+$ versus $\mathbb{N}_0 \ni 0$; the Lean formalization)**.** The original sources pose the problem for the positive integers: Davis, Entringer, Graham and Simmons ask whether a two-set partition of $\mathbb{Z}^+$ is possible, and Erdős–Graham (1979, p. 338) repeat the question, again for $\mathbb{Z}^+$. The Lean formalization in the DeepMind *formal-conjectures* repository (`FormalConjectures/ErdosProblems/197.lean`) instead partitions Lean's ground set $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, which *includes* 0 (and so differs from our $\mathbb{N} = \{1, 2, \dots\}$), and asks for bijections $f\colon \mathbb{N}_0 \simeq A$, $g\colon \mathbb{N}_0 \simeq B$ avoiding monotone 3-term APs — the bijection-from-$\mathbb{N}_0$ requirement being exactly our order-type-ω convention. The two formulations are not verbatim equivalent: a positive answer over $\mathbb{N}_0$ yields one over $\mathbb{Z}^+$ (delete the value 0; deletion preserves 3-AP-freeness), but conversely adjoining 0 to one part adds genuine constraints — the progressions $(0, d, 2d)$ with $d, 2d$ in that part — and we are not aware of a general argument that a permutable set stays permutable after adjoining 0. Our results are insensitive to the nuance: Theorem 21 shows S_{A} itself is not permutable, and since deletion preserves validity, $S_{\mathrm{A}} \cup \{0\}$ is not permutable either; hence the dyadic partition fails in both formulations, whichever part receives 0. (In the Lean encoding the values live in $\mathbb{N}_0$, so the common difference is a natural number and never negative; `HasMonotoneAP` requires strictly increasing indices whose value list equals $(a, a+d, a+2d)$ or its reverse, the reversed disjunct being the decreasing case. The predicate permits $d = 0$, but injectivity of the permutation rules that out, so the notion of "monotone 3-AP" agrees with ours.)

3 General obstructions

This section collects four restrictions that every admissible permutation of every set must obey. They are independent of the dyadic partition and are used freely later; the first is also the engine of Lemma 9.

3.1 The orbit obstruction

Lemma 4 (Orbit obstruction)**.** *Let S be permutable and $F \subseteq S$ finite. There is no infinite sequence $u_0 < u_1 < \cdots$ in S with $u_{k+1} = 2u_k - f_k$, $f_k \in F$.*

Proof. Let q be the largest position of an element of F in the permutation. Since the u_k are distinct, only finitely many occupy positions at most q; and since $u_k \to \infty$, eventually $u_k > \max F$. Choose K so that, for every $k \geq K$, $\mathrm{pos}(u_k) > q$ and $u_k > \max F$. Then $f_k < u_k$ and $\mathrm{pos}(f_k) \leq q < \mathrm{pos}(u_k)$, so the pair (f_k, u_k) is increasingly placed, and its completion $2u_k - f_k = u_{k+1}$ lies in S; were it placed after u_k, the triple (f_k, u_k, u_{k+1}) would be an increasing 3-AP. Hence $\mathrm{pos}(u_{k+1}) < \mathrm{pos}(u_k)$ for every $k \geq K$, an infinite strictly decreasing sequence of positions — impossible. $\square$

Taking $S = \mathbb{Z}^+$, $F = \{1, 2\}$ recovers the theorem of Davis–Entringer–Graham–Simmons (more precisely its monotone form; [1] in fact proves that an *increasing* 3-AP always occurs).

3.2 The balance law

Lemma 5 (Balanced placement)**.** *Let S be permutable via π, and consider the moment a value v is placed. Let L (resp. H) be the number of already-placed values in $[1, v)$ (resp. $(v, 2v)$). Then*

$$L - H \;\leq\; |S^c \cap (v, 2v)| \qquad \textit{and} \qquad H - L \;\leq\; |S^c \cap (0, v)|.$$

(The two directions are bounded by the codensities of S in different ranges; no symmetric single-range bound holds in general.[1])

Proof. Each placed $x < v$ forms an increasingly placed pair (x, v) with completion $2v - x \in (v, 2v)$, distinct for distinct x; each completion lying in S must already be placed. This gives $H \geq L - |S^c \cap (v, 2v)|$. In the other direction, each placed $u \in (v, 2v)$ forms a decreasingly placed pair (u, v) with completion $2v - u \in (0, v)$, distinct for distinct u, and each such completion lying in S must already be placed, whence $L \geq H - |S^c \cap (0, v)|$. $\square$

Corollary 6 (Records)**.** *When a left-to-right maximum v is placed, at most $|S^c \cap (v, 2v)|$ smaller values are already placed (apply the first inequality with $H = 0$).*

[1] We thank Jesse Geneson for correcting an earlier symmetric misstatement of this lemma.

3.3 Two absorption lemmas

Lemma 7 (Ratio-ascent obstruction). *Let $S \supseteq \{k, 2k, 3k, 5k\}$. No 3-AP-free arrangement of S places w before u for every pair $u \geq 2w$ (elements of S).*

Proof. The hypothesis forces $k \prec 2k$, $k \prec 3k$, $2k \prec 5k$. The progression $(k, 2k, 3k)$ with $k \prec 2k$, $k \prec 3k$ forces $3k \prec 2k$; then $3k \prec 2k \prec 5k$, and $(k, 3k, 5k)$ is an increasing 3-AP. □

Lemma 8 (van der Corput absorption; [10]). *Order the odd residues c modulo 2^k by the bit-reversal (van der Corput) rank of $(c-1)/2 \bmod 2^{k-1}$. Then for any two classes a ordered before b, the class $2b - a \bmod 2^k$ is ordered strictly before b.*

This is not new: it is Lemma 2.1 of [10] in different coordinates (via $c \mapsto (c-1)/2$). Explicitly, writing ρ_m for the bit-reversal rank modulo $m = 2^{k-1}$, the statement is the first equivalence $\rho_m(r) < \rho_m(r+\delta) \iff \rho_m(r+\delta) > \rho_m(r+2\delta)$ of that lemma, with $r = (a-1)/2$ and $\delta = (b-a)/2$. The underlying binary recursion goes back to [1] (p. 81) and to Ardal, Brown and Jungić [4]; the companion modular existence theorem for powers of two is due to Nathanson [5]. We include a short proof for completeness, and because the bit-reversal form is the one we use.

Proof. Write $\alpha = (a-1)/2$, $\beta = (b-1)/2$; the completion class has parameter $\zeta = 2\beta - \alpha \bmod 2^{k-1}$. If the lowest bits of α, β agree, then so does ζ's, and dividing by 2 preserves the form $\zeta' = 2\beta' - \alpha'$. At the first bit where they differ, the assumption $\mathrm{rev}(\alpha) < \mathrm{rev}(\beta)$ forces α's bit to be 0 and β's to be 1; since $\zeta \equiv \alpha$ at that bit, ζ's bit is 0, so $\mathrm{rev}(\zeta) < \mathrm{rev}(\beta)$ strictly. □

3.4 Partition rigidity

Lemma 9 (Ray piercing). *Let $\mathbb{Z}^+ = A \sqcup B$ with both parts permutable. Then for every $a \in A$ and every $d \geq 1$, the set $\{k \geq 0 : a + 2^k d \in B\}$ is infinite, and symmetrically with A and B exchanged.*

Proof. If $a + 2^k d \in A$ for all $k \geq k_0$, the values $u_j = a + 2^{k_0+j} d$ form an infinite orbit $u_{j+1} = 2u_j - a$ inside A with $F = \{a\}$, contradicting Lemma 4. □

Every affine doubling ray based in one team therefore meets the opposite team infinitely often (the precise content of the lemma — it does not assert literal alternation); partitions such as "even 2-adic valuation vs. odd" die immediately, and viable partitions are forced toward dyadic-interval alternation, motivating the canonical candidate S_{A}.

Remark 10 (Class-sink algebra). One algebraic fact about attacks within a residue class is worth recording, since it explains why powers of two are the natural moduli here. Fix $m = 2^k$ and a target class r. The attacker map $\varphi_r(b) = 2b - r$ on $\mathbb{Z}/m\mathbb{Z}$ is conjugate to doubling under $b \mapsto b - r$; hence every orbit eventually reaches the unique fixed point r. For a general modulus m the same conjugacy shows that the cycle structure of φ_r does not depend on r, so no class is distinguished.[2]

[2]We thank Jesse Geneson for correcting the general-m form of this statement; an earlier version asserted an r-dependent cycle structure.

4 Contiguous blocks cannot work

Definition 11. The *zone system* $Z(M)$: arrange the interval $(M, 2M]$ so that (a) no monotone 3-AP occurs, and (b) for every $y < z$ in $(M, 2M]$ with $2y - z \in (M/4, M/2]$, z precedes y.

Condition (b) is forced when the block $(M/4, M/2]$ is entirely placed before a contiguous run of $(M, 2M]$ within a team's sequence (its elements' completions $2y-x$ then must precede y).

Lemma 12 (Halving descent)**.** *If $Z(M)$ is satisfiable then so is $Z(\lfloor M/2 \rfloor)$.*

Proof. Restrict a witness to the even values of $(M, 2M]$ and halve; arithmetic progressions and the zone constraints scale exactly (the boundary cases are routine), and a superfluous top element may be deleted. □

Lemma 13 (Base cases; machine-checked)**.** *$Z(M)$ is unsatisfiable for $16 \leq M \leq 31$.*

This is a finite check on sixteen instances of at most 31 elements each, by exhaustive SAT with eager transitivity (script `e195_zone_bases.py`); the range is tight, since $Z(M)$ is *satisfiable* for $12 \leq M \leq 15$. Theorem 14 below is the only result depending on this verdict, and it is not used in the proof of the main theorem.

Theorem 14 (No contiguous-run solutions)**.** *In any 3-AP-free permutation of S_{A}, infinitely many blocks B_{2k} are not placed as contiguous runs.*

Proof. Lemmas 12 and 13 together make $Z(M)$ unsatisfiable for every $M \geq 16$ (halve repeatedly into $[16, 31]$). Suppose for contradiction that every block B_{2k} with $k \geq k_0 \geq 3$ is placed as a contiguous run. Contiguous runs are disjoint segments of the permutation, so for each $k > k_0$ one of the runs B_{2k-2}, B_{2k} entirely precedes the other. If B_{2k-2}'s run came first, then the whole block $(M/4, M/2]$, $M = 2^{2k-1}$, would be placed before the run of $(M, 2M]$, forcing condition (b) of the zone system as explained above — so the run of B_{2k} would witness $Z(M)$ with $M \geq 16$, which is unsatisfiable. Hence B_{2k}'s run precedes B_{2k-2}'s for every $k > k_0$, and the finish positions of the runs of $B_{2k_0}, B_{2k_0+2}, B_{2k_0+4}, \ldots$ form an infinite strictly decreasing sequence of positive integers — impossible. □

5 The order gadget and the reduction

The search described in the appendices compresses, for the canonical candidate S_{A}, into a one-scale-at-a-time statement about linear orders of single dyadic blocks. This section states that reduction precisely and proves it unconditionally — directly on a hypothetical permutation, using nothing from the appendices — and reports the machine verification of its finite instances; Section 6 then proves the required infeasibility by hand on the residue class $M \equiv 0 \pmod 8$ — which contains every dyadic scale — completing the proof of the main theorem (Theorem 21). Throughout, $b_j = M + j$ (*bottoms*) and $t_i = 2M - i$ (*tops*) denote elements of the interval $(M, 2M]$.

Definition 15 (Order gadget)**.** For an integer $M \geq 16$, the *order gadget* $\mathrm{OG}(M)$ asks for a linear order $\prec$ of the interval $(M, 2M]$ such that

(i) *(no monotone AP)* for every arithmetic progression $a < b < c$ inside $(M, 2M]$, neither $a \prec b \prec c$ nor $c \prec b \prec a$ holds; and

(ii) *(guards precede bottoms)* for $x \in \{15, 16\}$ and $1 \le j \le x/2$, the *guard* $t_{x-2j} = 2M+2j-x$ precedes the *bottom* $b_j = M + j$.

$\mathrm{OG}(M)$ is *infeasible* if no such order exists. (For $M \le 15$ some guard coincides with the bottom it protects — $t_{x-2j} = b_j$ exactly when $M = x - j$, e.g. $t_{14} = b_1$ at $M = 15$ — and the gadget degenerates; we never use those scales. For $16 \le M \le 22$ a guard can coincide with a *different* bottom, which is harmless: every attack pair is a genuine pair for all $M \ge 16$.)

Three pieces of boundary arithmetic, each a one-line computation (machine-checked exhaustively in the repository, script `e96_reduction_check.py`), make the gadget exactly the trace of the infinite problem on one block: for $x \in \{15, 16\}$ the completion $2b_j - x$ lies in $(M, 2M]$ precisely for $j \le x/2$, and then equals the guard t_{x-2j} (at $(x, j) = (16, 8)$ it is $2M$ itself, still inside the half-open block) — so (ii) lists *all* in-block completions of pairs (x, b_j) and nothing else. All other completions lie above $2M$; at the dyadic scales used in Theorem 16 they lie in the intervening odd dyadic block and hence outside S_{A}. Finally the guard–bottom pair's own downward completion $2b_j - t_{x-2j} = x$ is the attacking value itself, below the block, so the attack edges generate no unmodeled in-block constraint.

Theorem 16 (Order-gadget reduction; unconditional)**.** *If* $\mathrm{OG}(2^{2t-1})$ *is infeasible for infinitely many* $t \ge 4$*, then* S_{A} *is not* 3*-permutable.*

Proof. Suppose π is a 3-AP-free permutation of S_{A} and let $P = \max(\mathrm{pos}(15), \mathrm{pos}(16))$; note $15, 16 \in B_4 \subseteq S_{\mathrm{A}}$. Fix $t \ge 4$ and put $M = 2^{2t-1}$, so that $(M, 2M] = B_{2t} \subseteq S_{\mathrm{A}}$ and $15, 16 < M$.

We claim that some bottom b_j with $1 \le j \le 8$ has $\mathrm{pos}(b_j) \le P$. Suppose not, and order the block by position: $u \prec v :\iff \mathrm{pos}(u) < \mathrm{pos}(v)$. Constraint (i) is inherited from π. For (ii), take $x \in \{15, 16\}$ and $1 \le j \le x/2$. The pair (x, b_j) is placed increasingly ($\mathrm{pos}(x) \le P < \mathrm{pos}(b_j)$, $x < M < b_j$), and its completion $z = 2b_j - x = t_{x-2j}$ lies in $(M, 2M] \subseteq S_{\mathrm{A}}$ and differs from b_j (else $M = x - j \le 15$). If $\mathrm{pos}(z) > \mathrm{pos}(b_j)$, then (x, b_j, z) is a monotone 3-AP of π; hence $\mathrm{pos}(z) < \mathrm{pos}(b_j)$, i.e. $t_{x-2j} \prec b_j$. Thus $\prec$ witnesses $\mathrm{OG}(M)$, contradicting infeasibility.

So for every $t \ge 4$ with $\mathrm{OG}(2^{2t-1})$ infeasible, some element of $\{M+1, \dots, M+8\}$, $M = 2^{2t-1}$, occupies a position $\le P$. These blocks are disjoint, so infinitely many such t would place infinitely many distinct values among the first P positions — impossible. □

Remark 17 (Chunk form; the crowns cannot defend every block)**.** In the coordinates of Theorem 32 the identical argument reads: in any valid scheme s, for every $t \ge 4$ with $\mathrm{OG}(2^{2t-1})$ infeasible, some bottom value in $\{M+1, \dots, M+8\}$ of block B_{2t} has stage $\le \max(s(15), s(16))$. For if all eight had strictly larger stage, the induced order on the block (by stage, then fiber position) would satisfy $\mathrm{OG}(M)$: condition (i) holds for in-block triples directly by (A) and (B), and each attack $t_{x-2j} \prec b_j$ is forced whenever $s(x) < s(b_j)$ — by (A) the guard's stage cannot strictly exceed the bottom's, and at equal stages the forced-pair row $s(x) < s(y) = s(z) \Rightarrow z \prec y$ of (B) applies. The overflow step is then the finiteness of the fibers at stages $\le \max(s(15), s(16))$; *no* displacement normalization (Lemma 33) is needed anywhere in this route. This is the precise sense in which the two *crowns* 15 and 16 of Appendix B cannot defend every dyadic block: they would have to be placed below the bottom-eight of all but finitely many blocks. We use only the direction scheme ⇒ order; the converse (rebuilding a capped scheme from an OG witness, which would make the crown-cap infeasibility at a horizon *equivalent* to OG infeasibility at its top scale) has been verified only at the single scale $M = 128$, where the relevant family-MUS is single-block, and is not used here.

Proposition 18 (Machine verification; Cadical195)**.** OG(M) *is infeasible for every* M *with* $16 \le M \le 200$*, and for* $M = 512$*. In particular both tested scales of the dyadic family* $\{2^{2t-1}\}_{t\ge 4} = \{128, 512, 2048, \dots\}$ *are infeasible. The encodings assert (i) over all in-block triples and (ii) as unit clauses; order axioms are handled by lazy transitivity refinement, which is sound for infeasibility verdicts and was cross-validated against full eager* $O(n^3)$ *transitivity encodings at* $M = 40, 44$ *and — as an end-to-end recheck — at* $M = 128$ *(fresh eager instance,* 690,831 *clauses, UNSAT in about a second). The run at* $M = 2048$ *is unresolved (no verdict after* 200+ *refinement rounds), and the support growth reported below makes it unlikely that any finite sweep can settle the general statement.*

5.1 Attack cores: locating the load-bearing axioms

The hypothesis of Theorem 16 demands OG-infeasibility at infinitely many dyadic scales. The following machine facts locate a three-axiom core of the gadget that is already infeasible on exactly the residue class containing the dyadic family; Section 6 proves that core infeasibility by hand.

Proposition 19 (Attack cores; machine-checked)**.** *For every swept* M *(all of* $40 \le M \le 100$*, both parities, and* $M \in \{104, 108, 112, 120, 128, 150, 200\}$*), constraint (i) together with only the eleven attack units* $15\{1..7\} \cup 16\{1..4\}$ *— that is,* $t_{15-2j} \prec b_j$ *for* $j \le 7$ *and* $t_{16-2j} \prec b_j$ *for* $j \le 4$ *— is already infeasible. On residue subclasses, sharper cores hold, with the following verdicts at every swept scale (we make no claim outside the swept range): the four units* $\{t_{13} \prec b_1,\ t_{11} \prec b_2,\ t_5 \prec b_5,\ t_{10} \prec b_3\}$ *are infeasible with (i) precisely at the swept* $M \equiv 0 \pmod 4$*, and the three units*

$$\mathrm{C3} \;=\; \{\, t_5 \prec b_5, \quad t_3 \prec b_6, \quad t_{10} \prec b_3 \,\}$$

are infeasible with (i) precisely at the swept $M \equiv 0 \pmod 8$ *(satisfiable at every other swept residue, including* $M \equiv 4 \bmod 8$*). Only the infeasibility half of the last statement is used below, and it is proved unconditionally for all* $M \equiv 0 \pmod 8$ *in Theorem 22; the satisfiability half remains a computational observation.*

Every scale of the dyadic family is $\equiv 0 \pmod 8$, so Theorem 16 needs only the infeasibility of (i) $\wedge$ C3 on the class $M \equiv 0 \pmod 8$. That is exactly Theorem 22 below.

Remark 20 (What the experiments suggested about the shape of a proof)**.** Three recurring computational observations across the tested scales shaped the proof in Section 6. We record them because they explain the form the argument takes; none is a theorem, and nothing below depends on them. (1) Deletion-minimal refutation supports were observed to grow linearly with M (about 8 triples per unit of M over the swept range), which suggested that no fixed finite list of parametric AP-triples would certify infeasibility at all scales, and motivated looking for $\Theta(M)$-length mechanisms with an $O(1)$ description — the ladder floods below are of that form. (2) The forced relations behind the small-scale refutations were observed to be parity-locked, reversing at $M \not\equiv 0 \bmod 4$, which suggested that a uniform argument would have to stay inside a residue class; the dyadic family lives in $M \equiv 0 \pmod 8$. (3) Fixed-denominator band-limit windows of the constraint system were satisfiable at every scale tested, suggesting that the obstruction is genuinely asymptotic, so that a compactness or limit argument over order types was unlikely to reach it and a scale-uniform schema would be needed instead.

6 The C3 core theorem and the main theorem

Theorem 21 (Main theorem). $S_{\mathrm{A}} = \bigcup_{k\geq 2,\ k\ even} B_k = \bigcup_{k\geq 2,\ k\ even}(2^{k-1}, 2^k]$ *is not* 3-*permutable: it admits no permutation (of order type* ω*) free of monotone* 3*-term arithmetic progressions. In particular the canonical dyadic partition* $\mathbb{Z}^+ = S_{\mathrm{A}} \sqcup S_B$ *is not a solution of the Erdős–Graham two-set problem: any partition witnessing a YES answer to problem #197, if one exists, must differ from the dyadic one.*

Proof. Every scale of the dyadic family satisfies $M = 2^{2t-1} \equiv 0 \pmod 8$ and $M \geq 128$ for $t \geq 4$. By Theorem 22 below, the C3 core — hence a fortiori the full gadget $\mathrm{OG}(M)$, whose attack list (Definition 15(ii)) contains the three axioms of C3 as the instances $(x, j) = (15, 5), (15, 6), (16, 3)$, and whose constraint (i) is AP-freeness — is infeasible at every such scale. Theorem 16 applies. $\square$

The remainder of this section proves the core statement:

Theorem 22 (C3 core). *For every* $M \equiv 0 \pmod 8$ *with* $M \geq 16$*, no AP-free linear order of the interval* $(M, 2M]$ *satisfies the three axioms*

$$\mathrm{C3} = \{\, A_1\colon t_5 \prec b_5, \quad A_2\colon t_3 \prec b_6, \quad A_3\colon t_{10} \prec b_3 \,\}.$$

Here *AP-free* means condition (i) of Definition 15: no arithmetic progression $a < b < c$ inside the block occurs in monotone position order. The proof is a small toolkit of ladder lemmas (§6.1) followed by two forcing theorems (§6.2–§6.3). Every lemma application in the proofs has been machine-verified step by step at every $M \equiv 0 \pmod 4$ (resp. $\equiv 0 \pmod 8$) in $[12, 400]$ plus $M = 512, 1024$, and independently cross-validated; see Remark 29.

6.1 The ladder toolkit

AP-freeness is the *midpoint-extremal rule*: on every in-block AP (a, b, c), $c = 2b - a$, the midpoint b either precedes both endpoints or follows both. We use it as four unit rules (plus transitivity throughout):

R1: $a \prec b \Rightarrow c \prec b$, R3: $c \prec b \Rightarrow a \prec b$, R2: $b \prec c \Rightarrow b \prec a$, R4: $b \prec a \Rightarrow b \prec c$.

Write $m_0 = 3M/2$ (the arithmetic midpoint of the block: $b_j + t_j = 3M = 2m_0$ for all j, so every pair (b_j, t_j) mirrors through m_0). A *d-ladder* is a maximal run $w_0, w_1, \ldots$ of values of $(M, 2M]$ in arithmetic progression with difference d; the $d = 2$ ladders are the two parity classes, and the $d = 4$ ladders on odd values are *class A* $= \{v \equiv M + 1 \bmod 4\}$ and *class B* $= \{v \equiv M + 3 \bmod 4\}$.

Lemma 23 (Zigzag)**.** *Let $\prec$ be an AP-free linear order of an interval containing the d-ladder $w_0, \ldots, w_r$. Suppose $w_e \prec w_{e'}$ for some $|e - e'| = 1$. Then every rung w_i with $i \equiv e \pmod 2$ precedes each of its existing neighbors.*

Proof. Any three consecutive rungs form an AP with the middle rung as midpoint, so each rung either leads or trails both neighbors. The seed makes w_e a leader. If w_i leads, then from $w_i \prec w_{i+1}$, R1 on (w_i, w_{i+1}, w_{i+2}) gives $w_{i+2} \prec w_{i+1}$, and R4 on $(w_{i+1}, w_{i+2}, w_{i+3})$ gives $w_{i+2} \prec w_{i+3}$: w_{i+2} leads. Downward is the mirror argument. ☐

Lemma 24 (Phase dichotomy)**.** *Let $\prec$ be an AP-free linear order of a d-ladder $w_0, \ldots, w_r$ with $r \geq 1$. Then the ladder is globally in exactly one of its two zigzag phases: either all even-index rungs lead their existing neighbors, or all odd-index rungs do. Consequently the leader set of the ladder is one of its two half-classes modulo 2d, adjacent rungs strictly alternate leader/trailer, and a proof may case-split on the phase and detach any conclusion derived in both branches.*

Proof. The order orients the adjacent pair (w_0, w_1) one way or the other; apply Lemma 23 to that seed. ☐

Lemma 25 (Transfer lock)**.** *Let M be even, $M \geq 12$, and let $\prec$ be an AP-free linear order of $(M, 2M]$ (below $M = 12$ the four rungs named here collide, e.g. $t_5 = b_3$ at $M = 8$). On the odd $d = 2$ ladder $w_i = M + 1 + 2i$ $(0 \leq i \leq M/2 - 1)$ the four odd C3 values sit at $b_3 = w_1$, $b_5 = w_2$, $t_5 = w_{M/2-3}$, $t_3 = w_{M/2-2}$, and Lemma 23 locks the pair orientations together:*

$$M \equiv 0 \pmod 4 : \quad b_5 \prec b_3 \iff t_3 \prec t_5; \qquad M \equiv 2 \pmod 4 : \quad b_5 \prec b_3 \iff t_5 \prec t_3.$$

Proof. Each orientation of an adjacent pair seeds Lemma 23; read off whether the other pair's left member is a leader. At $M \equiv 0 \pmod 4$: $b_5 \prec b_3$ (i.e. $w_2 \prec w_1$) makes even indices leaders, and $M/2 - 2$ is even, so $t_3 = w_{M/2-2}$ leads its neighbor t_5. Conversely $t_3 \prec t_5$ seeds even-index leaders and $w_2 = b_5$ leads $w_1 = b_3$. The reverse orientations force the reverse conclusions, giving the biconditional; at $M \equiv 2 \pmod 4$ the parity of $M/2$ shifts the lock by one rung. ☐

Lemma 26 (Flood)**.** *Let $\prec$ be an AP-free linear order of $(M, 2M]$. Let C be the residue class $r \bmod g$ inside $(M, 2M]$, where $g \in \{2, 4\}$ (so C is a parity class for $g = 2$, and one of the two odd mod-4 classes for $g = 4$), and suppose the $d = g$ ladder of C is in a definite zigzag phase with leader set L. Let $c \in (M, 2M]$ be a center with $c \equiv r + g/2 \pmod g$, so that the mirror pairs $(c - e,\ c,\ c + e)$ with $e \equiv g/2 \pmod g$ have both members in C and form APs with midpoint c; call e admissible when both $c \pm e \in (M, 2M]$. Suppose some admissible e_0 carries a seed relation between c and a member $v \in \{c \pm e_0\}$. Then:*

- *(outward) if $c \prec v$, then $c \prec w$ for every $w \in C$ with $2c - w \in (M, 2M]$;*
- *(inward) if $v \prec c$, then $w \prec c$ for every such w.*

Moreover the conclusion holds in both zigzag phases of the C-ladder (phase-blindness), so by Lemma 24 it may be detached after a two-branch case split whenever the seed is available in both branches.

Proof. First, at each admissible e exactly one of $c+e$, $c-e$ is a leader: they differ by $2e \equiv g \pmod{2g}$, so they lie in the two different half-classes mod $2g$ of C, and L is one of them — whichever phase holds. Second, in a zigzag, adjacent rungs alternate and each leader leads both its neighbors.

Seed. From the relation at v, the mirror rule on the AP $(c-e_0, c, c+e_0)$ (R2/R4 outward, R1/R3 inward) gives the same-direction relation at the other member: both members of pair e_0 are related to c in the flood direction.

Outward, $e \to e+g$: let $x \in \{c \pm e\}$ be the leader of pair e. Its outward ladder-neighbor x' (same side, $|x'-c| = e+g$) satisfies $x \prec x'$; with $c \prec x$ known, transitivity gives $c \prec x'$, and the mirror rule floods $2c - x'$.

Outward, $e \to e-g$: let $w \in \{c \pm (e-g)\}$ be the trailer of pair $e-g$. Its outward neighbor w' (distance e, same side) is a leader and leads it: $w' \prec w$; with $c \prec w'$ known, $c \prec w$, and the mirror rule floods the other member.

Inward, $e \to e+g$: let x be the leader of pair $e+g$; it leads its inward neighbor x_{in} (distance e, same side): $x \prec x_{\text{in}} \prec c$, and the mirror rule floods the other member. *Inward,* $e \to e-g$: let x be the leader of pair $e-g$; it leads its outward neighbor (distance e): $x \prec x_{\text{out}} \prec c$; mirror floods the other member.

The admissible e form an interval of the residue class $g/2$ mod g, so the two inductions from e_0 reach every admissible e. Every step used only R1–R4 on genuine APs, transitivity, and zigzag edges of the given phase; the choice of side at each step is forced by the phase but *exists* in both phases. □

Three instances of Lemma 26 are used below. **POLAR** ($g = 2$, center m_0, class the odds; $M \equiv 0 \bmod 4$ makes m_0 even): seeded by the comparison of m_0 with t_5 (mirror b_5), it floods m_0 against *all* odd values (the mirror of b_l is t_l). The odd ladder's phase is pinned by the ambient hypotheses, so no dichotomy is needed. **P2-floods** ($g = 2$, an odd center c near m_0, class the evens): seeded by c vs. m_0 at $e_0 = |c - m_0|$ (supplied by POLAR); the even ladder's phase is unknown — phase dichotomy, both branches. **G4-floods** ($g = 4$, class A or B, center $c \equiv r+2 \pmod 4$ for that class's residue r): seeded at $e = 2$ by the odd ladder's zigzag edges between c and $c \pm 2$; the $d = 4$ ladder's phase is unknown — phase dichotomy, both branches.

The *mod-8 lock* of the whole problem sits in the center condition of the G4-floods: at $M \equiv 0 \pmod 8$ the two odd neighbors of m_0 satisfy $m_0 - 1 \equiv 3$ and $m_0 + 1 \equiv 1 \pmod 4$, so $m_0 - 1$ is a G4-center for class A and $m_0 + 1$ for class B. At $M \equiv 4 \pmod 8$ both congruences reverse: the two centers still exist, but with their class roles *swapped*, and the odd-ladder leader statuses at $m_0 \pm 1$ invert as well. Theorem 27 below survives this swap — it is stated and proved for all $M \equiv 0 \pmod 4$, absorbing the swap into the definition of the centers c^*, c^{**} — whereas Theorem 28 does not: there the two required seed sets cease to exist simultaneously, and the argument fails, as it must, since (i) $\wedge$ C3 is in fact *satisfiable* at $M \equiv 4 \pmod 8$ (Proposition 19). The mod-8 hypothesis of Theorem 22 therefore enters only through Theorem 28.

6.2 Layer 1: two axioms force the odd-pair orientations

Theorem 27 (L1)**.** *Let $M \equiv 0 \pmod 4$, $M \geq 12$, and let $\prec$ be an AP-free order of $(M, 2M]$ satisfying $A_2\colon t_3 \prec b_6$ and $A_3\colon t_{10} \prec b_3$. Then $b_5 \prec b_3$ and $t_3 \prec t_5$.*

Proof. It suffices to refute $S\colon b_3 \prec b_5$; given $b_5 \prec b_3$, Lemma 25 supplies $t_3 \prec t_5$. Assume A_2, A_3, S. By Lemma 23 (seed S) the odd ladder's leaders are the offsets $\equiv 3 \pmod 4$. Define

$$c^* = \begin{cases} m_0 + 1, & M \equiv 0 \pmod 8 \\ m_0 - 1, & M \equiv 4 \pmod 8 \end{cases} \qquad c^{**} = \begin{cases} m_0 - 1, & M \equiv 0 \pmod 8 \\ m_0 + 1, & M \equiv 4 \pmod 8 \end{cases}$$

so that $c^* \equiv 1 \pmod 4$ is a G4-center for class B whose odd neighbors $c^* \pm 2$ ($\equiv 3 \bmod 4$) are odd-ladder leaders, and $c^{**} \equiv 3 \pmod 4$ is a G4-center for class A and is itself an odd-ladder leader. Split on the comparison (m_0, t_5).

Case I: $t_5 \prec m_0$. POLAR-inward: every odd value $\prec m_0$. Then: (1) $b_3 \prec c^*$ by the G4-inward flood at c^* over class B — seeds $c^* \pm 2 \prec c^*$ (the leaders $c^* \pm 2$ lead their trailing neighbor c^*); $b_3 \in$ class B with mirror $2c^* - b_3 \in \{t_1, t_5\}$ in the block; both $d{=}4$ phases. (2) $c^* \prec t_{10}$ by the P2-outward flood at c^* over the evens — seed $c^* \prec m_0$ (POLAR, $e_0 = 1$); t_{10} even with mirror $2c^* - t_{10} \in \{M{+}8, M{+}12\}$ in the block; both even phases. (3) A_3: $t_{10} \prec b_3$. Together: the 3-cycle $b_3 \prec c^* \prec t_{10} \prec b_3$, a contradiction.

Case II: $m_0 \prec t_5$. POLAR-outward: $m_0 \prec$ every odd value. Then: (1) $c^{**} \prec t_3$ by the G4-outward flood at c^{**} over class A — seeds $c^{**} \prec c^{**} \pm 2$ (c^{**} is an odd-ladder leader); $t_3 \in$ class A with mirror $2c^{**} - t_3 \in \{b_1, b_5\}$; both phases. (2) $b_6 \prec c^{**}$ by the P2-inward flood at c^{**} over the evens — seed $m_0 \prec c^{**}$ (POLAR, $e_0 = 1$); b_6 even with mirror $2c^{**} - b_6 \in \{2M{-}8, 2M{-}4\}$; both phases. (3) A_2: $t_3 \prec b_6$. Together: the 3-cycle $c^{**} \prec t_3 \prec b_6 \prec c^{**}$, a contradiction.

Both cases are contradictory, so S is impossible. □

Note that Case I consumes only A_3 and Case II only A_2; the boundary arithmetic (mirrors in the block, seed admissibility) holds down to $M = 12$.

6.3 The flip: the third axiom is anti-forced

Theorem 28 (FLIP). *Let $M \equiv 0 \pmod 8$, $M \geq 16$, and let $\prec$ be an AP-free order of $(M, 2M]$ satisfying A_2, A_3 and $b_5 \prec b_3$. Then $t_5 \prec b_5$ is impossible: $b_5 \prec t_5$ ($= \neg A_1$) is forced.*

Proof. Assume $A_1\colon t_5 \prec b_5$ for contradiction. By Lemma 23 (seed $b_5 \prec b_3$) the odd ladder's leaders are the offsets $\equiv 1 \pmod 4$. Since $M \equiv 0 \pmod 8$: $m_0 - 1 \equiv 3 \pmod 4$ is a G4-center for class A whose odd neighbors $m_0 + 1$, $m_0 - 3$ (offsets $\equiv 1 \bmod 4$) are odd-ladder leaders; $m_0 + 1 \equiv 1 \pmod 4$ is a G4-center for class B and is itself an odd-ladder leader. Split on (m_0, t_5).

Case I: $t_5 \prec m_0$. POLAR-inward: every odd $\prec m_0$. Then: (1) $m_0 - 1 \prec t_{10}$ [P2-outward flood at $m_0 - 1$ over the evens; seed $m_0 - 1 \prec m_0$ (POLAR, $e_0 = 1$); mirror of t_{10} is $M + 8$; both phases]. (2) $m_0 - 1 \prec b_3$ [(1) $+A_3$]. (3) $m_0 - 1 \prec t_5$ [R4 on the AP (b_3, $m_0 - 1$, t_5): $b_3 + t_5 = 3M - 2 = 2(m_0 - 1)$]. (4) $m_0 - 1 \prec b_5$ [(3) $+A_1$]. (5) $b_5 \prec m_0 - 1$ [G4-inward flood at $m_0 - 1$ over class A; seeds $m_0 + 1 \prec m_0 - 1$ and $m_0 - 3 \prec m_0 - 1$ (leader edges); $b_5 \in$ class A at pair distance $M/2 - 6 \equiv 2 \pmod 4$ with mirror t_7 in the block; both phases]. (4) and (5) form a 2-cycle, a contradiction.

Case II: $m_0 \prec t_5$. POLAR-outward: $m_0 \prec$ every odd. Then: (1) $b_6 \prec m_0 + 1$ [P2-inward flood at $m_0 + 1$ over the evens; seed $m_0 \prec m_0 + 1$ (POLAR, $e_0 = 1$); mirror of b_6 is $2M - 4$; both phases]. (2) $t_3 \prec m_0 + 1$ [A_2 + (1)]. (3) $b_5 \prec m_0 + 1$ [R3 on the AP (b_5, $m_0 + 1$, t_3): $b_5 + t_3 = 3M + 2 = 2(m_0 + 1)$]. (4) $t_5 \prec m_0 + 1$ [A_1 + (3)]. (5) $m_0 + 1 \prec t_5$ [G4-outward flood at m_0+1 over class B; seeds $m_0+1 \prec m_0+3$ and $m_0+1 \prec m_0-1$ (m_0+1 is an odd-ladder leader); $t_5 \in$ class B at pair distance $M/2-6 \equiv 2 \pmod 4$ with mirror b_7; both phases]. (4) and (5) form a 2-cycle, a contradiction. □

Again Case I consumes A_3 and Case II consumes A_2, while A_1 is consumed symmetrically in both (step 4). The proof visibly fails at $M \equiv 4 \pmod 8$ in both cases: the centers $m_0 \mp 1$ land in the wrong mod-4 classes for their G4-floods and the odd-ladder leader statuses at $m_0 \pm 1$ invert, so neither seed set exists.

Proof of Theorem 22. $A_2 + A_3$ force $b_5 \prec b_3$ (Theorem 27; $M \equiv 0 \bmod 8 \subseteq 0 \bmod 4$). Then A_1 contradicts Theorem 28. □

Remark 29 (Machine verification of the schemas)**.** Although the proofs above are complete hand proofs, each is a *schema* whose instances at a given scale involve $\Theta(M)$ ladder steps; all have been machine-executed with strict per-step assertions (every AP's membership and arithmetic, every rule pattern, every leader/trailer and residue claim, every mirror bound, both branches of every phase dichotomy): Theorem 27 at every $M \equiv 0 \pmod 4$ in $[12, 400]$ plus $512, 1024$ (100 scales), Theorem 28 at every $M \equiv 0 \pmod 8$ in $[16, 400]$ plus $512, 1024$ (51 scales), and the $M \equiv 4 \pmod 8$ inapplicability at 35 scales (script `e113_c3_hand_proof.py`). An independent closure engine (plain R1–R4 + transitivity fixpoint, no knowledge of the schemas) refutes every branch of both case trees at 51 resp. 27 scales up to 512 (`e113b_closure_crossval.py`); and direct SAT checks of the two theorem *statements* at fresh scales never touched by the discovery loop ($M = 148, 212, 264, 328$ UNSAT; $M = 268, 332, \equiv 4 \bmod 8$, SAT) concur (`e114_theorem_spotcheck.py`). The earlier exhaustive base ledgers (AP + C3 UNSAT at every $M \equiv 0 \bmod 8$ in $[16, 256]$ and at 512) are now corroboration, not a dependency.

Remark 30 (What remains open)**.** The main theorem needs nothing beyond the class $M \equiv 0 \pmod 8$. The full order gadget is nevertheless conjectured infeasible at *every* $M \geq 16$ (machine-verified for $16 \leq M \leq 200$ and $M = 512$, Proposition 18); at residues $\not\equiv 0 \bmod 8$ the C3 core is satisfiable and other attack subsets take over (Proposition 19), so a proof would need per-residue analogues of Theorems 27–28 built from the same flood toolkit. This statement is not needed for Theorem 21. The Erdős–Graham problem #197 itself remains open: Theorem 21 eliminates the canonical candidate partition, but does not decide whether some other two-set partition works.

7 Machine verification and data availability

The proof chain of Theorem 21 — Theorem 16 followed by Theorem 22 — is human-readable throughout, and no step of it is delegated to a solver. Machine work enters in three distinct roles, which we separate explicitly because they carry very different evidential weight. (1) *Discovery*: the order gadget, the three-axiom core, and the mod-8 residue phenomenon were found by large-scale SAT experimentation; nothing in the final proof depends on how they were found. (2) *Audit of the writing*: Theorems 27 and 28 are $\forall M$ schemas, and each of their instances at a given scale unfolds into $\Theta(M)$ ladder steps; those instances have been executed with per-step assertions over a large finite range of scales by three independently written checking layers (Remark 29). This catches misstated lemmas and bookkeeping slips, and is emphatically not a substitute for the proofs, which are quantified over all $M \equiv 0 \pmod 8$). (3) *Standalone verdicts*: the finite claims of Lemma 13, Proposition 18, Proposition 19 and Appendix B are solver results, true at the scales stated and asserted at no others.

All experiment scripts, machine-audit checkers (including the mutation-test suite for the checkers themselves), solver logs, and witness data are contained in the accompanying repository, `https://github.com/Wkasel/erdos197`, together with `reproduce.sh`, which reruns the load-bearing verification stack for the main theorem (the schema audits of Remark 29, the closure-engine cross-validation, fresh direct SAT checks, re-emission and independent `drat-trim` checking of the DRAT certificates, and the reduction checks of Section 5) from a pinned environment. The repository contains DRAT certificates for exactly two instances, the C3-core refutations at $M = 128$ and $M = 512$; by Remark 29 these are corroboration rather than a dependency. The other machine verdicts quoted in the paper — among them the base cases of Lemma 13, the OG sweep of Proposition 18, the attack cores of Proposition 19, the machine atlas and the ladder rungs — are reproducible from the scripts but are not accompanied by proof certificates. The validity of the $\forall M$ schemas rests on the hand proofs of Sections 5–6 and not on any of these verdicts; their per-step machine audit (Remark 29) covers a finite range of scales only.

Acknowledgments and computational assistance

I thank Jesse Geneson for comments on an earlier version of this paper, which corrected two statements at the points where they apply: the balance law (Lemma 5), given there in a false symmetric form, and the class-sink observation (Remark 10), asserted there with an r-dependent cycle structure. His comments did not extend to a verification of Theorem 21 or of the chain leading to it.

This paper was prepared with machine assistance, which I disclose in full. The language model Claude (Anthropic) was used for proof auditing, code assistance, and editorial revision. The SAT solvers CaDiCaL and OR-Tools CP-SAT were used for the computational experiments reported in Section 7 and Appendix B, and for the discovery work described there. The author takes full responsibility for the contents of this paper, including every statement and proof and any error that remains.

A The chunk reduction

Every permutation of S_{A} of order type ω decomposes into consecutive finite segments (*chunks*); conversely, a choice of finite fibers and internal orders assembles into such a permutation. The following exact reformulation is the coordinate system for the elimination phase (Appendix B); as noted in the introduction, the reduction of Section 5 is proved without it, and nothing in this appendix is used in the proof of Theorem 21.

Definition 31. Let $S \subseteq \mathbb{Z}^+$ be infinite. A *stage function* for S is any $s\colon S \to \mathbb{N}$ with finite fibers. Given s and a linear order on each fiber, let T be the concatenation.

Theorem 32 (Chunk reduction; exact)**.** *T avoids monotone 3-term APs if and only if:*

(A) no AP triple (x, y, z), $z = 2y - x$, in S has strictly monotone stages ($s(x) < s(y) < s(z)$ or $s(z) < s(y) < s(x)$); and

(B) each fiber order contains the forced pairs

$$\begin{array}{ll} s(x) < s(y) = s(z) :\ z \prec y, & s(z) < s(y) = s(x) :\ x \prec y, \\ s(x) = s(y) < s(z) :\ y \prec x, & s(y) = s(z) < s(x) :\ y \prec z, \end{array}$$

and is non-monotone on every within-fiber triple. (Triples whose endpoint stages lie on one side of the middle's are automatically safe.)

Hence S is 3-permutable if and only if some stage function for S, together with a choice of fiber orders, satisfies (A) and (B). The fiber problems are finite and mutually independent given s. (Nothing in the statement or the proof uses any property of S; we apply it to $S = S_{\mathrm{A}}$ and, in Theorem 34, to $S = \mathbb{Z}^+$.)

Proof. Along positions of T the stage is non-decreasing, and within a stage the fiber order gives the positions. Suppose T avoids monotone 3-APs. If (A) failed at (x, y, z) with $s(x) < s(y) < s(z)$, then x, y, z occur in T in that position order with $x < y < z$ — an increasing 3-AP (the case $s(z) < s(y) < s(x)$ gives a decreasing one). If a forced pair of (B) failed, say $s(x) < s(y) = s(z)$ with $y \prec z$ in their fiber, then x, y, z occur in position order — again an AP; the other three rows and the within-fiber non-monotonicity requirement are identical. Conversely, suppose (A) and (B) hold and let (x, y, z), $z = 2y - x$, be placed monotonically in T, say $\mathrm{pos}(x) < \mathrm{pos}(y) < \mathrm{pos}(z)$ (the decreasing case is symmetric). Then $s(x) \le s(y) \le s(z)$. Strict inequalities throughout contradict (A); $s(x) = s(y) < s(z)$ puts $x \prec y$ in one fiber, contradicting the row $s(x) = s(y) < s(z) \Rightarrow y \prec x$ of (B); $s(x) < s(y) = s(z)$ puts $y \prec z$, contradicting $z \prec y$; and $s(x) = s(y) = s(z)$ is a monotone within-fiber triple, excluded by (B). Triples whose endpoint stages lie on one side of the middle's stage can never be placed monotonically, since the middle's position is then extremal among the three; this is the safety clause. The full case table over all stage patterns and fiber orders has additionally been machine-checked exhaustively (`e96_reduction_check.py`, part P2). □

Lemma 33 (Normalization: running-max chunking)**.** *Every 3-AP-free permutation π of S_{A} (of order type ω) admits a valid scheme — a stage function with finite fibers satisfying (A)∧(B), whose concatenation is π itself — with $s(v) \ge \mathrm{block}(v)/2$ for every v.*

Proof. Set $s(v) = \max_{q \le \mathrm{pos}(v)} \mathrm{block}(\pi(q))/2$ (the running maximum of half-block-indices). Along positions s is non-decreasing, so its fibers are consecutive segments of π and the concatenation (with the fiber orders induced by π) is π; Theorem 32 then gives (A)∧(B). Since $\mathrm{block}(v)/2$ enters the maximum at v's own position, $s(v) \ge \mathrm{block}(v)/2$. Each fiber is finite: π is onto S_{A}, so a value of block index $> 2\sigma$ occurs at some finite position, after which the running maximum exceeds σ. □

The point of Lemma 33 is that non-negative *displacement* (next section) is a harmless normalization, not an automatic property: an arbitrary chunk decomposition can place a high-block value in an early chunk. Restrictions of normalized schemes stay normalized, which is what the divergence criterion below consumes.

Theorem 34 (DEGS via chunks)**.** *$\mathbb{Z}^+$ itself is not 3-permutable.*

Proof. Suppose $s\colon \mathbb{Z}^+ \to \mathbb{N}$ satisfies (A) with finite fibers and put $\sigma = s(1)$. For every y with $s(y) > \sigma$, the triple $(1, y, 2y - 1)$ forces $s(2y - 1) \le s(y)$ by (A). The orbit y_0, $y_{i+1} = 2y_i - 1$

is strictly increasing; only finitely many of its members can have stage $\leq \sigma$ (those fibers are finite), after which its stages are non-increasing, hence eventually confined to one finite fiber — impossible for an infinite set. $\square$

This recovers the Davis–Entringer–Graham–Simmons theorem (in its monotone form; [1] proves the stronger increasing-3-AP statement) in the chunk coordinates and isolates what a permutable team must do: break every such reflected orbit out of the team. The dyadic team S_{A} breaks single-reflector orbits in two steps (completions land in odd blocks); the question is whether the multi-scale accounting below can be survived.

Lemma 35 (Same block)**.** *In every AP triple of S_{A} the middle y and top z lie in the same even block. ($z < 2y \leq 2^{k+1}$ and block $k+1$ is odd.)*

Lemma 36 (Escape)**.** *If $y \geq 3 \cdot 4^{s-1}$ (top quarter of block $2s$) and $x \leq 4^{s-1}$, then $2y - x > 4^s$: completions of (lower value, top-quarter value) pairs land in the odd block above — out of team.*

Lemma 37 (Bottom-half characterization)**.** *Deferring $D_k \subseteq$ block k by a uniform delay (adding a constant to s on D_k) creates no violation of condition (A) if $D_k \subseteq (2^{k-1}, 3 \cdot 2^{k-2}]$: the witness midpoints $y = (x + z)/2$ of a bottom-half $z \in D_k$ against an S_{A}-value x in a lower block (so $x \leq 2^{k-2}$) satisfy $2^{k-2} < y \leq 2^{k-1}$, i.e. lie in the odd block below. Conversely a deferred top-half z has genuine in-team witnesses, forcing its witness midpoints (bottom-quarter values) to defer with it.*

B Computational observations and conjectures

Everything in this appendix is a record of *finite-scale computation*: measurements, solver verdicts at particular horizons, and the conjectures they suggest. None of it is used in the proof of Theorem 21, whose chain is Theorem 16 together with Theorem 22. We report it because it is what located the order gadget and the two crowns 15, 16, and because the conjectures it raises seem to us the natural next questions. Statements here that are quantified over all scales are explicitly labelled as conjectures; the rest are verdicts at the horizons named, and carry no proof certificates (see Section 7).

B.1 Local realizability and the tower characterization

Definition 38 (Pure-complete systems)**.** For an S_{A}-block top X, the *pure-complete-X system* asks for an arrangement of $S_{\mathrm{A}} \cap [1, X]$ with no monotone 3-AP. (All completions of in-range pairs lie in $(X, 2X]$, an odd block, or inside the range; hence this is exactly the restriction of the infinite problem.)

Lemma 39 (Horizon collapse)**.** *Deleting any set of values from a valid arrangement preserves validity; hence auxiliary values above X never aid the pure-complete-X system.*

Theorem 40 (Tower characterization)**.** *S_{A} is permutable iff there exists $B \colon \mathbb{N} \to \mathbb{N}$ such that for every block top X the system [pure-complete-X and every v has at most $B(v)$ predecessors] is satisfiable.*

Proof. (⇒) restrict a permutation and take $B(v)$ its positions. (⇐) the restriction sets are finite and nested; Kőnig's lemma yields a consistent tower whose limit has each v preceded by at most $B(v)$ values, hence order type ω; all constraints are finitary and inherited. □

Computationally: pure-complete-X is satisfiable for $X \le 1024$ (explicit witnesses in the repository), and the completion-pressure function $g_X(L) = \min \max_{v \le L} \mathrm{pos}(v)$ satisfies $g_{256}(64) = 153$, this value being optimal. The equality $g_{256}(64) = 153$ means that at least $153 - 42 = 111$ of the 128 values in $(128, 256]$ precede the last-placed member of $S_{\mathrm{A}} \cap [1, 64]$ (recall $|S_{\mathrm{A}} \cap [1, 64]| = 42$, so at most 42 of the first 153 positions can be occupied by members of $S_{\mathrm{A}} \cap [1, 64]$ itself). In the optimal arrangements found, those early values of $(128, 256]$ were concentrated in a single residue class modulo 8. Divergence of $g_X(L)$ in X for fixed L would prove non-permutability directly; our measurements did not exhibit divergence (the subset-tolerance is near-total), and this route was abandoned in favour of the one taken in Sections 5–6. There is no tension with Theorem 21: stabilization of $g_X(L)$ for each fixed L does not by itself produce a single bound B as required by the tower characterization, because the minimizing arrangements vary with L.

B.2 Self-similar witnesses, fragility, and pumping obstructions

Adding the exact scale-invariance $u \prec w \iff 4u \prec 4w$ to the pure-complete systems remains satisfiable at $X = 256$ and $X = 1024$ (independently audited witnesses); however the audited 1024-witness does not extend to a self-similar 4096-arrangement (solver verdict, uncertified), so the self-similar tower also has dead branches.

We also record a calibration fact: requiring, in addition, that the ± 1 neighbors (within the same block and team) of each forced completion also precede the pair's later element ("radius-1 robustness") is infeasible in every setting we tested — including plain intervals $[1, n]$ from $n = 8$, where no partition is involved at all. We have no proof that this persists, but within the tested range radius-1 robustness fails for reasons that have nothing to do with the dyadic partition, so it does not discriminate between partitions and is of no use as a satisfiability criterion; a pumping argument would instead have to manage rounding drift on the thin family of far-pair constraints only. The radius-one experiments are *evidence* against placement rules in which position varies slowly with value; we neither formalize such a class of rules here nor prove a theorem about it, and we draw no further inference from the observation.

B.3 The negative atlas

Theorem 41 (Block-granular death)**.** *No stage function whose fibers are unions of whole even blocks satisfies (A)∧(B).*

Proof. For an even $b \ge 6$ define the *fatal core* F_b: the variables are the relative order of the elements of the block B_b, and the constraints are

(a) the forced pair $z \prec y$ for every AP triple (x, y, z), $z = 2y - x$, with $x \in B_{b-2}$ and $y, z \in B_b$ — these are the instances of the row $s(x) < s(y) = s(z) \Rightarrow z \prec y$ of (B) that a boundary between B_{b-2} and B_b creates; and

(b) non-monotonicity on every AP triple lying inside B_b.

Any fiber whose lowest block is B_b contains F_b as a subsystem, and a supersystem of an unsatisfiable system is unsatisfiable.

F_6 is unsatisfiable: it has 51 forced pairs and 240 triples on the 32 elements of B_6, and is refuted by CaDiCaL in well under a second (script `e59`, independently reproduced by `e67`; this is a solver verdict on one finite instance and carries no proof certificate, cf. Section 7). The map $v \mapsto 4v$ then embeds F_b into F_{b+2}: it preserves S_{A}, sends block j to block $j{+}2$, and preserves AP triples; so every F_b with $b \geq 6$ even is unsatisfiable. It remains to see that a block-granular scheme has an *ascending* fiber boundary above block 4, i.e. some $b \geq 6$ with $s(B_{b-2}) < s(B_b)$ (this is what puts the forced pairs of F_b into the system). If there were none, then $s(B_4) \geq s(B_6) \geq s(B_8) \geq \cdots$ would be a non-increasing sequence of natural numbers, hence eventually constant; infinitely many blocks would then share a single fiber, contradicting the finiteness of fibers. $\square$

Theorem 42 (×4 restriction; window death is permanent)**.** *For the window class* $\{s : s(v) - \mathrm{block}(v)/2 \in [0, w]\}$*: feasibility at horizon* $4N$ *implies feasibility at horizon* N *(restrict along* $v \mapsto 4v$ *and shift stages by one). Consequently the machine verdicts below are permanent (hold at all larger horizons), and an infinite solution in a window class would imply feasibility at every finite horizon.*

Machine atlas (CP-SAT and CDCL cross-checks; solver logs and witnesses in the repository, but no proof certificates for the negative verdicts): bottom-half relief of depth 1 and depth ≤ 2 infeasible at $N = 256$ (and 1024); **window-2 infeasible at** $N = 1024$ (CP-SAT, 746s; independently CDCL/unary-ladder, 1760s) — hence, by Theorem 42, infeasible at every horizon ≥ 1024, so there is no infinite window-2 solution. (At the smaller horizon $N = 256$ window 2 is still feasible, but only via whole-block merges, of minimum total displacement 41; the merged skeleton is $C_s = 2^s \cdot [3 \cdot 2^{s-2}, 2^s]$, the top-quarter multiples of 2^s — precisely the set singled out by Lemma 36.)

Among the classes of stage function we were able to search, the atlas leaves one shape of candidate for a YES: stage functions with unbounded displacement (chunks mixing stragglers of unboundedly many blocks). Closing the remaining gap — between Theorems 41/42 and general finite-fiber schemes — by the per-block deferral calculus of Lemma 37 was the state of the elimination phase before Section 5, where the gap is closed instead by the order-gadget route, which does not use the chunk calculus at all. The same analysis applies to S_B by the evident mirror symmetry (checked by machine at 4096).

B.4 The displacement ladder

For a stage function s write $\delta(v) = s(v) - \mathrm{block}(v)/2$ (*displacement*); a scheme is *normalized* if $\delta \geq 0$ everywhere, which by Lemma 33 is no loss of generality for schemes induced by genuine permutations. Define

$$L(m) \;=\; \min_{\text{normalized schemes at horizon } 4^m} \; \max_{v \in S_{\mathrm{A}},\; v \leq 16} \delta(v),$$

the minimum over all normalized finite-fiber schemes satisfying (A)∧(B) at horizon 4^m (no bound on any other value's displacement); the maximum is over the ten values $S_{\mathrm{A}} \cap [1, 16] = \{3, 4\} \cup \{9, \ldots, 16\}$.

Proposition 43 (Ladder rungs; machine-checked). *$L(3) \le 1$; $L(4) \ge 2$ (infeasibility of the cap $\delta \le 1$ on $S_{\mathrm{A}} \cap [1, 16]$ at $N = 256$, verified independently by CP-SAT, 252s, and CDCL with a unary-ladder encoding, 15s); $L(5) = 3$ (lower bound: CDCL, 2304s, a single run; upper bound: a window-3 witness at $N = 1024$, 995s — which also gives $L(4) \le 3$, since L is non-decreasing in m). Moreover the minimal infeasible cap-coalition at $N = 256$ is exactly $\{15, 16\}$ (deletion-minimal): every scheme has $\delta(15) \ge 2$ or $\delta(16) \ge 2$ — the crowns of the two competing class towers ($15 = 1111_2$, apex of the $\equiv -1$ skeleton; $16 = 10000_2$, the block top, apex of the $\equiv 0$ skeleton).*

Theorem 44 (Divergence criterion). *If $L(m) \to \infty$ then S_{A} is not 3-permutable.*

Proof. An infinite valid permutation induces (Lemma 33) a *normalized* valid scheme, and by restriction a normalized valid scheme at every horizon 4^m (completions of values $\le 4^m$ that lie beyond the horizon fall in the odd block $(4^m, 2 \cdot 4^m]$, out of team), and the ten displacements $\delta(v)$, $v \in S_{\mathrm{A}} \cap [1, 16]$, are constants of the infinite scheme. For m with $L(m) > \max_{v \in S_{\mathrm{A}},\, v \le 16} \delta(v)$ this is a contradiction, by the exactness of Theorem 32. $\square$

The conjectured growth $L(m) = m - 2$ matches the window-death schedule (window w dying at horizon 4^{w+3}) and the observed per-block descent of displacement in minimal solutions (δ profile $(m-2, m-3, \ldots, 1, 0)$ across blocks $4, 6, \ldots, 2m$). The missing step is a self-propagating squeeze $L(m+1) \ge L(m) + 1$; the $\times 4$ restriction alone yields only the diagonal transfer $L_{b+1}(m+1) \ge L_b(m)$ for the capped set $[1, 4^b]$.

References

[1] J. A. Davis, R. C. Entringer, R. L. Graham and G. J. Simmons, *On permutations containing no long arithmetic progressions*, Acta Arith. **34** (1977), 81–90.

[2] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory: van der Waerden's theorem and related topics*, Enseign. Math. (2) **25** (1979), 325–344.

[3] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L'Enseignement Mathématique **28**, Genève, 1980.

[4] H. Ardal, T. Brown and V. Jungić, *Chaotic orderings of the rationals and reals*, Amer. Math. Monthly **118** (2011), 921–925.

[5] M. B. Nathanson, *Permutations, periodicity, and chaos*, J. Combin. Theory Ser. A **22** (1977), 61–68.

[6] T. D. LeSaulnier and S. Vijay, *On permutations avoiding arithmetic progressions*, Discrete Math. **311** (2011), 205–207.

[7] J. Geneson, *Forbidden arithmetic progressions in permutations of subsets of the integers*, Discrete Math. **342** (2019), 1489–1491; arXiv:1803.06334.

[8] S. Adenwalla, *Avoiding monotone arithmetic progressions in permutations of integers*, Discrete Math. **347** (2024), 114183; arXiv:2211.04451.

[9] M. Hirose and S. Saito, *Characterization of order structures avoiding three-term arithmetic progressions*, Order **42** (2025), 231–239; arXiv:2404.13510.

[10] J. Geneson, *Density bounds for permutations avoiding monotone arithmetic progressions*, arXiv:2608.12604 (2026).

[11] B. S. Ho, *3AP-free permutations have no exponential growth rate*, arXiv:2602.13617 (2026).

[12] T. F. Bloom, *Erdős Problem #197*, `https://www.erdosproblems.com/197`, accessed 2026-08-25.

[13] Google DeepMind, *Formal Conjectures*: `FormalConjectures/ErdosProblems/197.lean`, `https://github.com/google-deepmind/formal-conjectures`.